Software Requirements Specification

Online Exam Portal - Release 1

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For team review; not yet approved

This specification defines the required behavior of Online Exam Portal Release 1 for development, review and validation. It follows the supplied Jackfruit SRS structure and maps 17 scope requirements to the corrected Jira backlog. Requirements describe planned behavior; implementation evidence, executed tests and approval remain pending.

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Revision history

Version	Date	Author	Change
1.0	2 October 2026	Guntupalli Manaswi	Initial Release 1 requirements
1.1	3 October 2026	Guntupalli Manaswi	Three-member metadata; guideline structure, use cases, traceability and references revised

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1 Introduction

1.1 Purpose

Specify the complete Release 1 portal behavior for a web-based MCQ examination workflow. This SRS is the baseline for acceptance criteria, design and later validation. The team shall review the requirements before treating version 1.1 as approved.

1.2 Intended audience

Developers use Sections 3 to 7 to implement interfaces, validation and domain behavior. Testers derive positive, negative, boundary and concurrency checks from each uniquely identified requirement. Instructors and students review the workflow and role rules. Faculty evaluators review scope, traceability and individual contribution. The project plan, architecture, detailed design and validation specification are maintained as separate deliverables.

1.3 Product scope

The portal enables instructors to prepare and publish exam-specific MCQs, students to complete timed attempts, and the server to finalize, evaluate and store results. Release 1 includes role-based authentication, exam and MCQ management, publication, online answer persistence, navigation, timed manual and timeout submission, automatic scoring and restricted result viewing. The minimum Admin workflow provisions and manages accounts and roles.

Excluded: Negative marking; AI question generation or proctoring; reusable question banks; course or enrollment management; advanced analytics; CSV export; offline synchronization; mobile applications.

No course or enrollment subsystem, public self-registration, retake subsystem or answer-key disclosure is required. The goal is a dependable course-project workflow, not a claim of production certification or proctoring.

1.4 References

• JACKFRUIT Software Engineering Project Guidelines 2026, UE24CS341A, supplied Word document, reviewed 3 October 2026. This source defines the proposal, SRS, planning, testing and reporting formats.

• Online Exam Portal Release 1 problem statement and scope supplied by Guntupalli Manaswi, 2 October 2026.

• OEP-16 Project Proposal / Synopsis, version 1.1, 3 October 2026; repository working branch docs/OEP-16-project-proposal.

• Jira OEP project, stories OEP-14 OEP-15 OEP-24 through OEP-31 OEP-37 through OEP-42; current acceptance criteria read 2 October 2026. Access: https://tanushaprakash.atlassian.net/browse/OEP-17.

• Project source and document repository: https://github.com/Manaswi-Guntupalli/SE_MINI_PROJECT.

2 Overall description

2.1 Product perspective

The system is a new standalone web application. A React browser client communicates with an Express application on Node.js; MongoDB stores users, exams, questions, attempts, answers and results. The server is authoritative for identity, authorization, availability, deadlines, finalization and scoring. A server expiry process handles attempts whose browsers are inactive. GitHub and Jira are development tools, not runtime dependencies for taking an exam.

2.2 Product functions

Authenticate and authorize users; administer accounts; create and manage owned draft exams and MCQs; publish immutable valid content; list exams; start or resume one student attempt; save and revisit answers; enforce timing; finalize manually or on expiry; evaluate and display authorized results.

2.3 User classes and characteristics

Role	Characteristics and permitted workflow
Student	Basic browser skills; uses listing attempt and own-result screens. Cannot author exams or view another student’s attempt.
Instructor	Prepares assessments and reviews results; manages only owned exams. Cannot administer accounts.
Admin	Trusted account administrator; provisions users and assigns a single role and active status. Does not inherit exam or student permissions.

2.4 Operating environment

Internet-connected desktop or laptop browser with JavaScript and cookies enabled; current stable Chrome, Edge or Firefox at validation time. Development may run on Windows, Linux or macOS. The application server runs a supported Node.js release with Express; MongoDB provides durable storage. Exact dependency versions and hosting configuration are to be pinned in lockfiles and the architecture document before implementation validation.

2.5 Design and implementation constraints

Use MERN and Agile delivery. Enforce security on the server, not only through UI hiding. Use MongoDB constraints and atomic updates for uniqueness and answer/finalization races. Server-managed sessions shall support production HttpOnly and Secure cookies, a documented SameSite policy and CSRF protection. Use Git contributions, PR review, GitHub Actions CI/CD, Docker and static/security validation.

2.6 Assumptions and dependencies

Users have stable online access; offline synchronization is excluded. An initial Admin is provisioned securely. Institution-specific account identifiers, retention policy, final hosting details and exact versions await confirmation. Appendix B and the dedicated RTM workbook are planned under OEP-18 and OEP-19; no executed tests or code references are available yet.

3 External interface requirements

3.1 User interfaces

Screen	Required visible behavior
Login	Login identifier password submit action and safe failure messages; logout is available after authentication.
Admin accounts	List accounts; create account; assign a single permitted role; change active status; show validation errors without exposing passwords.
Instructor dashboard	Owned exams with title draft/published status duration and optional start time; permitted management and results actions.
Exam editor	Draft settings and question/option operations; correct-option selection and actionable publication validation.
Student exam list	Published exams with eligibility and Start Resume or Submitted status; no drafts or answer keys.
Attempt	Question and options; saved selection; previous/next actions; remaining time; pending/saved/failed save indicator; explicit Submit.
Results	Student own score/maximum or instructor owned-exam student identities and scores; pending evaluation empty loading and error states.

Keyboard users shall be able to log in, choose an answer, navigate and submit using labeled controls. Errors shall identify the relevant field or action, avoid raw database details and never imply an unsuccessful answer save succeeded. Confirm destructive draft deletion and manual submission; unanswered questions do not prevent submission. Timezone labels shall accompany displayed availability times.

3.2 Software interfaces

React exchanges JSON with Express over HTTP in local development and HTTPS in deployed use. Requests carry validated exam, question and option identifiers; session identity comes from a protected cookie. API responses expose only role-permitted fields. Express accesses MongoDB through Mongoose or another documented supported driver; database errors are mapped to safe application errors. Detailed endpoint names, payloads and indexes belong to OEP-22 and OEP-43.

3.3 Communications interfaces

Use same-origin browser/API deployment where feasible. If separate origins are used, permit only documented origins with credential-aware CORS. Protect state-changing requests from CSRF. Transmit timestamps as UTC ISO 8601 values and return server time with the deadline. Enforce authorization on every protected request. Do not transmit passwords in URLs, answer keys in attempt responses or student-supplied scores.

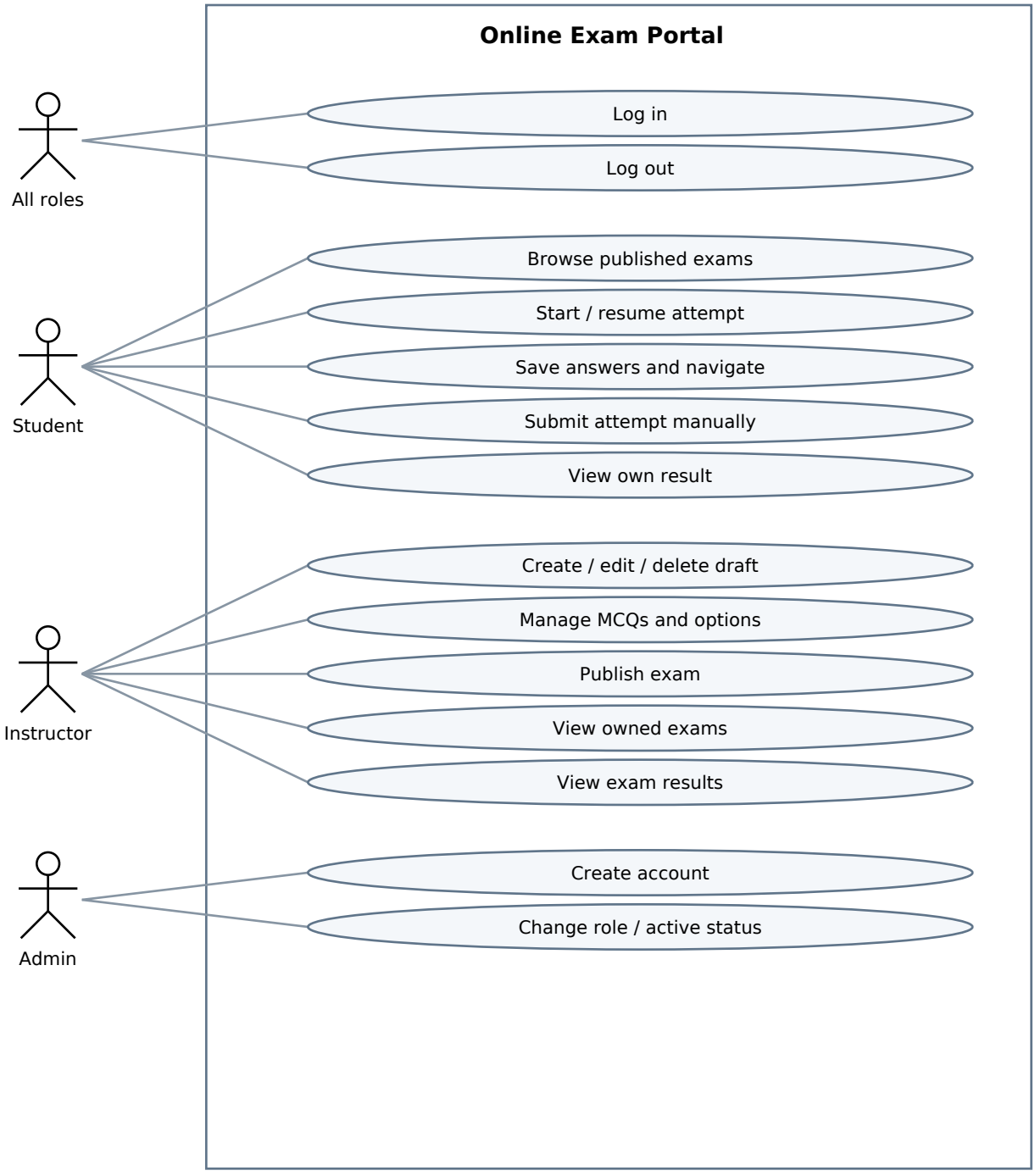
3.4 Hardware interfaces

No dedicated hardware, camera, sensors, offline client or mobile application is required. The browser, server and database need ordinary network and storage resources; deployment sizing is defined later.

4 Analysis models

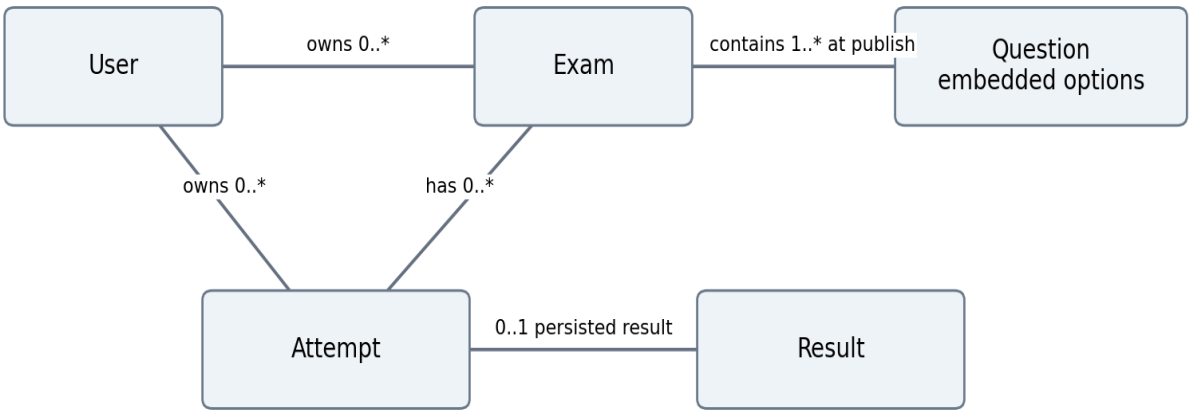
4.1 Use case model

All three actors authenticate and log out. The role-specific use cases below describe the permitted application workflows. Publication validation is part of Publish; evaluation follows both manual and timeout finalization.



4.2 Conceptual data model

A User owns exams when acting as Instructor and owns attempts when acting as Student. Each exam contains questions with their own options. A published exam has one or more questions. Each attempt references one exam and one student, stores one latest saved response per question and yields at most one persisted result. Question/option storage may be embedded or referenced in MongoDB; the logical membership rules remain the same.



Unique constraints: login identifier; Student plus Exam attempt; Attempt plus Question response when responses are separate records; Attempt result. User role/status changes preserve historic references. The diagrams are analysis models, not a claim of an implemented schema.

5 System features and functional requirements

The following 17 primary requirement IDs decompose the supplied scope for traceability. “Shall” specifies required planned behavior. Detailed checks within each ID are acceptance conditions. Every feature gives a stimulus and expected response, including invalid-input and unauthorized cases.

5.1 Authentication and account permissions

Priority High. Stimulus: a user submits credentials, logs out, or invokes a protected action; an Admin creates or changes an account. Response: authenticate only valid active accounts and enforce current role and ownership on each request.

FR 01 Authentication

Requirement ID FR-01

The system shall authenticate valid credentials for an active account and reject invalid or inactive credentials with a safe error. It shall hash passwords through an established password-hashing library, never store or log plaintext passwords, and use one documented server-managed session mechanism. Logout shall invalidate the server session and clear the browser cookie. Reusing a logged-out session shall fail. Anonymous protected requests shall be rejected.

FR 02 Role enforcement and account administration

Requirement ID FR-02

The server shall enforce exactly one Student, Instructor or Admin role and resource ownership on every protected operation. Only Admins shall provision accounts with a unique validated login identifier and update role or active status. Duplicate or invalid accounts shall be rejected without partial creation. Current stored role/status shall take effect on subsequent requests in existing sessions. Students and Instructors shall not change roles or create privileged accounts. Historical associations shall remain intact; secure initial Admin setup shall use no committed default credential. Admin status alone shall not grant instructor or student workflows.

5.2 Draft exam management

Priority High. Stimulus: an Instructor creates, edits or deletes an exam. Response: persist only validated owned draft changes; reject writes to published or foreign exams.

FR 03 Create an exam

Requirement ID FR-03

An authenticated Instructor shall create an owned draft with a nonblank title, optional description, optional availability start time and finite positive duration in documented units. The server shall validate all inputs before persistence, store the owner from session identity and display the draft on the owner dashboard. Store timestamps consistently and display timezone. Drafts shall not be visible or startable by Students.

FR 04 Edit and delete drafts

Requirement ID FR-04

Only the owner shall edit or delete a draft exam. Edits shall apply FR 03 validation; deletion shall remove or safely handle associated questions/options with no orphaned active content. Published exams, questions, options and answer keys shall not be edited or deleted. Missing, invalid, non-owner or published writes shall cause a safe error and no unauthorized change. Management operations shall preserve historic attempts/results.

5.3 MCQ management and publication

Priority High. Stimulus: the owner changes questions/options or requests Publish. Response: validate membership and content; persist draft operations or freeze a valid published definition.

FR 05 Manage MCQ questions

Requirement ID FR-05

Only the owner of a draft exam shall add, edit and delete its questions with nonblank text. Each question shall belong to that exam. Drafts may be incomplete; incomplete definitions shall not publish. Deleting a question shall safely remove or handle its options. Published question changes shall fail.

FR 06 Manage options and correct selection

Requirement ID FR-06

Only the draft owner shall add, edit, replace or delete options and change the correct selection. Option text shall be nonblank. A publishable question shall have at least two options and exactly one correct option belonging to that question. Removing the selected correct option shall require another correct selection before publication. Foreign option identifiers and multiple correct selections shall be rejected or leave the draft in an explicitly incomplete, unpublishable state. No reusable question bank is included.

FR 07 Publish an exam

Requirement ID FR-07

Only the owner shall publish a draft whose title/duration are valid, which contains at least one question and whose questions satisfy FR 05 and FR 06. Invalid publication shall return actionable validation details and retain draft status. Valid publication shall persist published status and freeze settings, questions, options and the answer key. Repeated publication shall not duplicate the exam. The optional availability start time shall control start eligibility on the server.

5.4 Exam discovery

Priority Medium for owner listing and High for student discovery. Stimulus: an Instructor or Student opens a dashboard. Response: return only role-permitted exams and permitted actions.

FR 08 View owned examinations

Requirement ID FR-08

The Instructor dashboard shall list only owned exams with title, status, duration and configured start time, and provide working create, draft edit/delete, MCQ, publication and results actions where permitted. Guessed foreign identifiers shall not reveal management data. The interface shall define loading, empty and failure states and reflect persisted changes after reload.

FR 09 View published examinations

Requirement ID FR-09

The Student list shall show published exams with title, duration, start time and current eligibility/attempt status, without drafts or answer keys. It shall enable Start only after the configured gate, show Resume for an active attempt and prevent restarting a submitted one. Starting shall revalidate eligibility on the server. Empty, loading and failure states shall be explicit; course enrollment shall not be a dependency.

5.5 Timed student attempts

Priority High. Stimulus: a Student starts or resumes an exam, selects an answer, moves between questions or reloads. Response: return the same authorized persisted attempt, maintain its deadline and safely store valid answers.

FR 10 Start and resume one attempt

Requirement ID FR-10

The server shall verify Student role, published status and start-time eligibility. It shall persist one attempt per Student and Exam with owner, exam, server start time, deadline equal to start plus duration, and active status before delivering content. Repeated or concurrent Start requests shall return the same active attempt without resetting the deadline. Reload/resume shall restore saved answers and deadline. A submitted attempt shall not restart; an expired active attempt shall be finalized. Attempt content shall omit correct-option identifiers, correctness flags and answer keys; other students’ attempts shall be denied.

FR 11 Enforce the timer

Requirement ID FR-11

The client shall display remaining time from the stored server deadline with server-time synchronization. Device clock changes, reload and resume shall not extend the deadline. Server writes received at or after deadline shall fail. At expiry the UI shall stop answer editing and request timeout finalization; independent server processing shall handle closed browsers. Timing/API errors shall not reset the attempt.

FR 12 Save answers and navigate

Requirement ID FR-12

Students shall view the current question/options and navigate previous and next questions in their own active attempt. Selecting or changing an option shall persist one response per attempt/question; returning or reloading shall restore the saved selection. Unanswered questions shall remain unanswered. The server shall validate question/exam and option/question membership and require active status and server time strictly before deadline. Updates shall be serialized or versioned so a delayed request cannot overwrite a newer acknowledged selection. Pending, saved and failed indicators shall be accurate; retry shall be possible only while time remains. Submitted, expired, foreign and unauthorized answer writes shall fail without changing data.

Deadline boundary example

For a 30-minute exam started at server time 10:00:00 UTC, the persisted deadline is 10:30:00 UTC. A valid answer received at 10:29:59 may save; one received at 10:30:00 shall be rejected. The availability start gate does not replace the per-attempt deadline. The example illustrates required semantics, not an executed test.

5.6 Manual and automatic submission

Priority High. Stimulus: a Student selects Submit or the server detects expiry. Response: freeze one consistent saved response set through the same conditional active-to-submitted transition and start evaluation.

FR 13 Submit manually

Requirement ID FR-13

The Student shall explicitly submit their own active attempt. Before successful manual finalization, the system shall complete pending saves while time permits, or clearly state that only successfully saved answers can be submitted. It shall not silently claim failed saves were persisted. The server shall atomically finalize the attempt once, persist final status and submission time and invoke evaluation. Repeated submissions shall return the same stored outcome. Concurrent answer/submit requests shall not modify the frozen response set after finalization.

FR 14 Submit automatically on timeout

Requirement ID FR-14

At server deadline the attempt shall become ineligible for further writes and be finalized with the last successfully saved answers; unanswered questions remain unanswered. A documented server expiry process shall run at a configured interval and finalize even with a closed browser or stopped JavaScript. Manual and timeout requests shall share the same race-safe transition and shall not duplicate submission. Finalization shall complete within the configured processing interval during healthy server operation; after restart, overdue attempts shall be recovered without granting more time. The interval and recovery checks shall be recorded in the design and validation specification.

5.7 Evaluation and result viewing

Priority High. Stimulus: a manual or timeout submission finalizes, or an authorized user opens results. Response: calculate and persist a reproducible result or explicitly show evaluation pending.

FR 15 Automatic evaluation and score calculation

Requirement ID FR-15

Only finalized attempts shall be evaluated against the immutable published answer key. Each correct response shall earn 1 point; wrong or unanswered responses earn 0. Score shall equal their sum and maximum shall equal question count, with 0 <= score <= maximum. The system shall persist score, maximum, attempt association and evaluated status. Student-supplied scores/correctness shall not be accepted. Concurrent or repeated evaluation shall not duplicate results or add points twice. On failure, the submitted response set shall remain durable and evaluation shall be retried by the server without reopening answers.

FR 16 Student result viewing

Requirement ID FR-16

Students shall see only their own submitted/evaluated result with exam identity, stored score and maximum. Pending evaluation, unavailable results, loading and failure states shall be clear. Direct access to another student’s result shall fail without leakage. Reload shall show the same persisted result. Answer-key disclosure is not required.

FR 17 Instructor result viewing

Requirement ID FR-17

Instructors shall select an owned exam and view student identity, stored score and maximum for that exam, distinguishing submitted/evaluation-pending attempts. Foreign-exam result access shall fail without leakage. No-results, loading and failure states shall be defined. Reload shall reflect persisted results. Advanced analytics and CSV export are excluded.

6 Other nonfunctional requirements

The security and correctness requirements below are mandatory. Numerical performance/usability targets are proposed acceptance baselines for team review, not measured results or guaranteed production capacity. The validation plan shall record environment, datasets and outcomes.

6.1 Performance requirements

NFR-P01 Proposed baseline: with 25 concurrent users, up to 50 questions per exam, on the documented validation environment, at least 95% of login, listing, answer-save and result-read requests shall complete within 2 seconds, excluding intentionally injected failures. Measure over a 10-minute representative run. This is a validation workload, not a new hard product limit. Record hardware/network conditions and approve or revise the baseline before load testing.

NFR-P02 The displayed timer shall update at least once per second during foreground browser operation. Server deadline enforcement remains authoritative irrespective of client refresh rate. Expiry processing shall satisfy FR 14 using the configured interval; the design shall state the actual interval and recovery behavior.

6.2 Safety and data integrity

NFR-S01 Acknowledged saved responses, finalized submissions and results shall survive page reload and a normal application process restart. On storage failure the server shall report failure rather than acknowledge an unpersisted save. NFR-S02 Destructive operations shall apply only to owned drafts; published definitions and historic associations shall remain intact. The product does not control physical safety equipment; no safety certification is claimed.

6.3 Security requirements

• NFR-SEC01 Hash passwords; protect production session cookies with HttpOnly Secure and documented SameSite settings; invalidate logout sessions and protect state-changing requests against CSRF.

• NFR-SEC02 Enforce current roles, active status and ownership for every protected request. Test direct unauthorized identifiers, not only hidden UI controls.

• NFR-SEC03 Validate types, lengths and membership on the server; allowlist accepted fields and reject score/correctness manipulation and invalid MongoDB query input. Use HTTPS in deployed use and documented CORS if origins differ.

• NFR-SEC04 Exclude passwords, session secrets, answer keys and sensitive request bodies from logs and unauthorized responses. Keep credentials in environment configuration outside Git.

• NFR-SEC05 Verify valid and forbidden actions for all three roles, session reuse after logout, account deactivation, CSRF and input validation through recorded security checks. No certification or completed penetration test is claimed.

6.4 Software quality attributes

NFR-Q01 Reliability and correctness: one attempt, one frozen submission and one result per permitted identity relationship; test repeated requests, deadline boundaries, stale saves and concurrent transitions. NFR-Q02 Usability: complete the Student login/start/answer/navigation/submit workflow by keyboard; identify all actionable controls and save failures. NFR-Q03 Maintainability and testability: separate UI, API, authorization, domain logic and persistence; review changes and run automated unit/integration checks in CI. Generate code and branch coverage reports; thresholds await team approval. NFR-Q04 Portability: verify the workflow in the selected current Chrome Edge and Firefox versions and record the exact versions at execution.

6.5 Business rules and domain requirements

Rule	Required principle
BR-01	Each account has exactly one role; Admin manages users but does not inherit other workflows.
BR-02	Instructors author only owned drafts. Published settings content and answer keys are immutable.
BR-03	Every publishable question has nonblank text at least two nonblank options and exactly one correct option belonging to it.
BR-04	A Student may start one attempt per exam after the optional start gate; resume preserves its original deadline.
BR-05	The server deadline is start time plus duration; answer writes require server time strictly before it and active status.
BR-06	Finalization freezes acknowledged saved responses exactly once. Unanswered responses earn zero.
BR-07	Correct earns 1; wrong/unanswered earns 0; maximum equals question count. No negative marking.
BR-08	Students see their own results; Instructors see only their owned-exam results.
BR-09	Role/status changes preserve historic associations and affect subsequent authorization checks.
BR-10	Internet access is required for saves; offline synchronization and retakes are excluded.

6.5.1 Attempt lifecycle

State	Permitted behavior and next state
Active before deadline	Answer writes and navigation allowed. Manual submission transitions atomically to Submitted.
Active at or after deadline	No answer writes. Expiry processing transitions to Submitted; elapsed time is never extended.
Submitted evaluation pending	No edits or restart. Evaluate or retry using the frozen saved responses.
Evaluated	Persisted result is readable by authorized users; repeated evaluation returns the same outcome.

7 Other requirements

DATA-01 Persist users, owned exams, questions/options, attempts, answers and results with valid references and uniqueness constraints. DATA-02 Store times consistently in UTC and present timezone-aware labels. DATA-03 Account deactivation shall preserve records. Institution-specific retention/deletion policy is pending; do not introduce destructive historic deletion by assumption.

PROC-01 Use Agile Jira stories, keyed Git branches/commits/PRs, teammate code review, GitHub Actions, Docker and static analysis as planned. Every team member shall implement and explain a complete functional contribution. PROC-02 The validation specification shall plan 8–10 unit/integration/system test cases per implemented use case, including valid/invalid data and precise expected results; actual outcomes and pass/fail remain Pending until executed. Record code/branch coverage, security checks and traceability as required by the course.

Appendix A Glossary

Term	Meaning
MCQ	Multiple-choice question with one correct option in this release.
Draft	Unpublished editable exam owned by its Instructor.
Published	Validated immutable exam definition; start eligibility also depends on availability time.
Attempt	A Student’s unique persisted exam session with start time deadline and answers.
Deadline	Server-calculated last permissible time boundary; writes at or after it fail.
Acknowledged answer	A selection the server has successfully persisted and confirmed.
Finalization	Atomic transition that freezes the saved response set and records submission.
Evaluation pending	Submitted attempt whose durable score has not yet been successfully calculated.
MERN	MongoDB Express React and Node.js.
SRS RTM	Software Requirements Specification; Requirement Traceability Matrix.
CSRF	Cross-site request forgery; requires protection for state-changing session requests.
CI CD	Continuous integration and continuous deployment.

Appendix B Field layout and report requirements

The detailed Excel field-layout workbook is the separate OEP-18 deliverable scheduled for 4 October. Its initial baseline shall describe planned fields and constraints; implemented-model evidence remains Pending. The following entity and report inventory is provided to make this SRS reviewable, without claiming the workbook is already complete.

Entity or report	Required information
User	Unique login identifier password hash name role active status; no plaintext stored password.
Exam and MCQ	Owner title description duration optional availability start status; question text option identifiers/text and private correct-option reference.
Attempt and answer	Student exam start deadline status submission time; question selected option and update/version information where needed.
Result	Unique attempt association score maximum evaluated status and evaluation time.
Student result view	Exam identity score maximum and pending/unavailable state.
Instructor results view	Owned exam identity student identity score maximum and pending evaluation state.

Companion workbook: docs/OEP-18_Field_Layout.xlsx (planned, OEP-18; 4 October 2026). The Excel deliverable shall include Field, Length, Data Type, Description, Is Mandatory and report fields. The inventory above is supplementary; the workbook must be supplied before the complete Phase-1 package is considered finished.

Appendix C Initial requirement traceability

The matrix includes every primary functional requirement and the guideline evidence columns. Pending means the referenced artifact or executed test evidence is not yet available. Jira mapping is supplied separately below; initial requirement traceability does not certify implementation.

Sl. No.	Requirement ID	Brief description	Architecture reference	Design reference	Code file reference	Test case ID	System test case ID
1	FR-01	Authentication	Pending	Pending	Pending	Pending	Pending
2	FR-02	Roles and Admin accounts	Pending	Pending	Pending	Pending	Pending
3	FR-03	Create exam	Pending	Pending	Pending	Pending	Pending
4	FR-04	Edit/delete drafts	Pending	Pending	Pending	Pending	Pending
5	FR-05	Manage questions	Pending	Pending	Pending	Pending	Pending
6	FR-06	Manage options/correct choice	Pending	Pending	Pending	Pending	Pending
7	FR-07	Publish exam	Pending	Pending	Pending	Pending	Pending
8	FR-08	Owned exam dashboard	Pending	Pending	Pending	Pending	Pending
9	FR-09	Student published exam list	Pending	Pending	Pending	Pending	Pending
10	FR-10	Start/resume one attempt	Pending	Pending	Pending	Pending	Pending
11	FR-11	Timer enforcement	Pending	Pending	Pending	Pending	Pending
12	FR-12	Save answers/navigation	Pending	Pending	Pending	Pending	Pending
13	FR-13	Manual submission	Pending	Pending	Pending	Pending	Pending
14	FR-14	Timeout submission	Pending	Pending	Pending	Pending	Pending
15	FR-15	Evaluate and calculate score	Pending	Pending	Pending	Pending	Pending
16	FR-16	Student own result	Pending	Pending	Pending	Pending	Pending
17	FR-17	Instructor owned-exam results	Pending	Pending	Pending	Pending	Pending

Requirement ID	Jira implementing stories
FR-01	OEP-14
FR-02	OEP-15 OEP-42
FR-03	OEP-24
FR-04	OEP-25

Requirement ID	Jira implementing stories
FR-05	OEP-26
FR-06	OEP-26
FR-07	OEP-27
FR-08	OEP-28
FR-09	OEP-29
FR-10	OEP-30
FR-11	OEP-37
FR-12	OEP-31
FR-13	OEP-38
FR-14	OEP-37 OEP-38
FR-15	OEP-39
FR-16	OEP-40
FR-17	OEP-41

Validation focus: FR-01/02 authentication and forbidden access; FR-03–09 ownership and publication validation; FR-10–14 resume, stale saves, deadline boundary and closed-browser/racing submission; FR-15 all-correct/all-wrong/mixed/unanswered scoring and retries; FR-16/17 visibility and pending results. Design detailed test IDs under OEP-23; execute and record evidence under OEP-44. NFR and business-rule IDs shall also be added to the dedicated RTM.